INCH-POUND
MIL-DTL-62006C(AT)
1 July 1998
SUPERSEDING
MIL-PRF-62006B(AT)
23 October 1995

DETAIL SPECIFICATION

COOLER, OIL: TRANSMISSION AND ENGINE

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers oil cooler assemblies for air cooling engine and transmission oil (see 6.1), and individual cores which make up the assembly types.

1.2 Classification. The types and classes of oil coolers are as follows (see 6.2 and 6.5):

Type I	- Engine and transmission oil cooler end cores.
Class A	- Engine only (see drawing 7003292).
Class B	- Transmission only (see drawing 7003291).
Class C	- Transmission only (see drawing 7985070).
Type II	- Engine and transmission oil cooler side cores.
Class D	- Engine and transmission (see drawing 7003293).
Class E	- Engine and transmission (see drawing 7003294).
Type III	- Engine and transmission oil cooler assemblies.
Class F	- Engine and transmission (see drawing 7003290).
Class G	- Engine and transmission (see drawing 8698621).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000 by using Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2930

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-2104	- Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.

FEDERAL

TT-P-1757	- Primer Coating, Alkyd Base, One Component.

(Unless otherwise indicated, copies of the above specifications are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS

DEPARTMENT OF DEFENSE

7003290	- Cooler Assembly, Engine and Transmission Oil.
7003291	- Core, Oil Cooler, End.
7003292	- Core, Oil Cooler, End.
7003293	- Core, Oil Cooler, Side.
7003294	- Core, Oil Cooler, Side.
7985070	- Core, Oil Cooler.
8698621	- Cooler Assembly, Engine and Transmission Oil.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D235	- Standard Specification for Mineral Spirits (Petroleum Spirits) (Hydrocarbon Dry Cleaning Solvent) (DoD Adopted).

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

GENERAL MOTORS CORPORATION (GM)

GM 9540P	- Accelerated Corrosion Test.

(Copies of GM publications may be obtained from General Motors Corporation, c/o Global Engineering, 15 Inverness Way, Englewood, CO 80112.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (se 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design and construction. Cooler assemblies and cores shall be in accordance with the applicable drawings as specified in table I.

TABLE I. Applicable drawings.

Classification (see 1.2)	Drawings	Component used on	
		Engine	Transmission
Type I			
Class A	7003292	X	
Class B	7003291		X
Class C	7985070		X
Type II			
Class D	7003293	X	X
Class E	7003294	X	X
Type III			
Class F	7003290	X	X
Class G	8698621	X	X

3.2.1 Welding, brazing, and soldering. Welding, brazing, and soldering shall be in accordance with the manufacturer's standard practices. Cooler assemblies and cores shall be free of flux after performance of fusion operations.

3.3 Materials. Unless otherwise specified herein, materials shall be as specified on applicable drawings (see table I).

3.3.1 Corrosion resistance. Corrosion protection of all exposed exterior surfaces shall be equal to or exceed that provided by treatment with a chemical conversion coating in accordance with type 1A of MIL-C-5541, followed by application of a primer coating in accordance with TT-P-1757.

3.3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Operating requirements. Cooler assemblies and cores shall provide the following functional, operational, and performance capabilities.

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3.4.1 Leakage. Cooler assemblies and cores shall not leak when subject to 400 pounds per square inch (psi) of hydrostatic pressure for at least 10 minutes, and 15 000 hydraulic pressure cycles from 0 to 300 psi.

3.4.2 Heat rejection. Heat rejection for the cooler assemblies and cores shall be not less than as specified in table II.

TABLE II. Heat rejection for engine and transmission cooler cores and assemblies.

Type (see 1.2)	Class (see 1.2)	Heat rejection-British thermal units (Btu) (kiloJoules/minute)	
		Engine	Transmission
I	A	2729 (2880)	---
	B & C	---	6484 (6840)
II	D	1354 (1430)	1643 (1735)
	E	1266 (1335)	1563 (1650)
III	F & G	5280 (5570)	9525 (10050)

3.5 Support and ownership requirements. Cooler assemblies and cores shall possess the following life cycle ownership characteristics.

3.5.1 Safety. Cooler assemblies and cores shall not produce any hazards to personnel or the environment resulting from the use of asbestos, cadmium, radioactive materials, toxic chemicals, or other hazardous substances.

3.5.2 Contamination resistance. Cooler assemblies and cores shall withstand a jarring drainage of mineral spirits without evidence of loose particles.

3.5.3 Identification marking. Unless otherwise specified (see 6.2), the cooler assemblies and cores shall be marked with the National Stock Number (NSN), the manufacturer's identification code (CAGE), and the part or identifying number (PIN) (see 6.5). Marking shall be permanent and legible.

3.6 Operating environment requirements. Cooler assemblies and cores shall withstand vibration through the frequency range as specified in figures 1 and 2.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.1.1).
b. Conformance inspection (see 4.1.2).
c. Acceptance tests (see 4.1.3).

4.1.1 First article inspection. Unless otherwise specified (see 6.2 and 6.3), first article inspection shall be performed on three cooler assemblies or cores. The inspection shall include all the examinations and tests as specified in table III.

4.1.1.1 Initial production inspection. Unless otherwise specified (see 6.2), two cooler assemblies or cores from the first 100 cooler assemblies or cores produced shall be selected for initial production testing. This inspection shall include the verifications as specified in 4.3.3, 4.4.1, and 4.4.2.

TABLE III. Verification methods.

Title	Require-ments	Verifica-tion	Title	Require-ments	Verifica-tion
Design and construction	3.2	4.3.3	**Support and ownership requirements**	3.5	4.5
Welding, brazing, and soldering	3.2.1	4.3.3.1	Safety	3.5.1	4.5.1
Materials	3.3	4.3.3	Contamination resistance	3.5.2	4.5.2
Corrosion resistance	3.3.1	4.3.3.2	Identification marking	3.5.3	4.5.3
Operating requirements	3.4	4.4	**Operating environment requirements**	3.6	4.6
Leakage	3.4.1	4.4.1			
Heat rejection	3.4.2	4.4.2	Vibration	3.6	4.6

4.1.2 Conformance inspection. Conformance inspection shall include the verifications listed in table III as specified in the contract (see 6.2 and 6.4).

4.1.3 Acceptance tests (100 percent (%)). Each cooler assembly and/or core shall be subjected to and pass the tests as specified in 4.4.1.

4.1.4 Sampling. The sampling plan shall be as specified in the contract (see 6.2).

4.2 Order of inspection. The inspection sequence may be in any order, except that the verifications specified in 4.3.3.1 and 4.6 shall precede the testing of all the operating requirements (see 4.4).

4.3 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously approved or previous qualified designs.

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4.3.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures, to verify performance. See the contract for alternatives that replace verification required by this specification.

4.3.2 Inspection conditions. Unless otherwise specified (see 6.2), ambient air temperature shall be 75 ± 10 degrees Fahrenheit (°F) (24 ± 5.5 degrees Celsius (°C)), barometric pressure shall be 28.54 (+2, -3) inches (in.) mercury (96.1 + 6.7/-10.1 kilopascals (kPa)), and relative humidity shall be 50 ± 30 %.

4.3.3 Design, construction, and materials verification. Use any of the methods outlined in 4.3 to assure that cooler assemblies and/or cores conform to the applicable drawings as specified in table I.

4.3.3.1 Welding, brazing, and soldering verification. Circulate water at 180 to 200°F (82 to 93°C) through the core for 10 minutes. Following the water circulation, the chloride content shall not exceed 40 parts per million above existing level for local water.

4.3.3.2 Corrosion resistance verification. Cooler assemblies and/or cores shall be tested in accordance with GM9540P, method A, duration D, and show no evidence of corrosion. All operating requirements verifications (see 4.4) shall be passed after exposure to this test.

4.4 Operating requirements verifications. Complete each verification under 4.4.

4.4.1 Leakage tests. Perform the tests as specified in 4.4.1.1 and 4.4.1.2.

4.4.1.1 Pneumatic pressure test. Cooler assemblies and/or cores shall be submerged in water at a temperature of 120 to 140°F (49 to 60°C). While submerged, the internal pressure shall be gradually applied, using dry air or nitrogen, until a pressure of 400 psi (2758 kPa) is achieved and maintained for a period of no less than 10 minutes. There shall be no evidence of leakage or permanent distortion of the cooler.

4.4.1.2 Hydraulic pressure test. Cores shall be subjected to 15 000 hydraulic pressure cycles from 0 to 300 psi (2068 kPa), at a rate of 13 to 20 cycles per min, at an oil temperature of 170 to 200°F (77 to 93°C), and on a test stand capable of maintaining the specified pressure, cycle rate, and temperature. Oil used for the test shall be 40 grade oil, conforming to MIL-PRF-2104, or equivalent. There shall be no evidence of leakage.

4.4.2 Heat rejection tests. Heat rejection tests shall be conducted using oil of the grade indicated entering the cooler assemblies and/or cores at the oil flow rates, air inlet and outlet temperatures, and air flow rates listed in table IV. Verify that heat rejection is within specified limits.

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TABLE IV. Air and oil properties used in determining heat rejection.

Type, class (component)	Air				Oil			
	Flow 1/ lb/min (kg/min)	Static drop 1/ in. H_20 (kPa)	Inlet tempera-ture °F (°C)	Grade (SAE)	Flow 1/ gal/min (L/min)	Pressure drop psi (kPa)	Inlet tempera-ture °F (°C)	Outlet tempera-ture °F (°C)
I, A (engine)	170 (77)	11 (2.7)	140 (60)	15W-40	150 (567)	30 (210)	---	300 (149)
B and C (trans-mission)	275 (125)	7.5 (1.9)	140 (60)	10W	284 (1074)	30 (210)	225 (107)	---
II, D (engine)	89 (40)	11 (2.7)	140 (60)	15W-40	80 (302)	30 (210)	---	300 (149)
(trans-mission)	69 (31)	7.5 (1.9)	140 (60)	10W	78 (295)	25 (170)	225 (107)	---
II, E (engine)	84 (38)	11 (2.7)	140 (60)	15W-40	80 (302)	30 (210)	---	300 (149)
(trans-mission)	65 (29)	7.5 (1.9)	140 (60)	10W	78 (295)	25 (170)	225 (107)	---
III, F & G (engine)	900 (408)	11 (2.7)	140 (60)	15W-40	310 (1172)	30 (210)	---	300 (149)
(trans-mission)	900 (408)	7.5 (1.9)	140 (60)	10W	440 (1663)	25 (170)	225 (107)	---

1/ The following previously undefined abbreviations are used in this table: kg/min = kilograms per minute, lb = pounds, and H_20=water.

4.5 Support and ownership requirements verification. Complete each verification under 4.5.

4.5.1 Hazardous material evaluation. Use one or more of the methods outlined in 4.3 to verify that safety hazards are not present.

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4.5.2 Contamination resistance verification. Cooler assemblies and/or cores shall be half-filled with mineral spirits per ASTM D235, or equivalent, then capped and shaken in both horizontal and vertical planes. Upon draining through both ports, no particles shall be visible on Whitman #41, or equivalent, filter paper.

4.5.3 Identification marking evaluation. Visually inspect coolers and cores for properly, legibly, and permanently marked NSN, CAGE, and PIN numbers (see 6.5).

4.6 Operating environment requirement verification. The vibration test procedure shall be conducted as follows, and shall subsequently be subjected to and pass the tests as specified under 4.4:

a. Test time at temperature of 80°F ± 10°F (27°C ± 5°C) shall be a total of 9 hours with 3 hours in each of the three mutually perpendicular axes of the cooler assemblies and/or cores.
b. The cooler assemblies and/or cores shall be soft mounted to the test fixture at six points. The cooler assemblies and/or cores shall be filled with water and oriented as it is installed in the vehicle.
c. The input control sensing device(s) shall be rigidly attached to the vibration platform as near as possible to the attachment point(s) of the test item. Additional vibration sensors shall be located on the test cooler assemblies and/or cores to determine resonant frequencies and amplification factors. Locations to be selected should include the main structure and large component parts. The sensor sizes and weights shall be limited so that their effect on the dynamic responses being measured is minimal. For sinusoidal vibration, when necessary for obtaining uniform results, a tracking filter should be used in the vibration exciter control feedback loop prior to the servo input.
d. A search shall be conducted to determine resonant frequencies along each major axis. These frequencies shall be determined by varying the frequency of applied vibration slowly through the frequency range specified on figure 1 at reduced test levels but with sufficient amplitude to excite the cooler assemblies and/or cores.
e. The cooler assemblies and/or cores shall be vibrated along each axis at the determined resonant frequencies. If more than three resonances are found in any one axis, the three most severe are to be chosen. Vibration shall be performed for 30 minutes in accordance with the test level specified in figure 1. Remaining time is to be utilized in the vibration sweeps test.
f. The vibration sweeps test shall be in accordance with the test level specified in figure 1. The time schedule shall be 15 minutes for sweep frequency range of 5 to 500 to 5 Hertz (Hz). When cooler assembly and/or core resonances below 5 Hz are measured, the test curves shall be extended to 2 to 500 to 2 Hz and the sweep

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time shall be 18 minutes. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 2. The specified sweep time is that of an ascending plus a descending sweep time and is twice the ascending sweep time shown on figure 2 for the specified range.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cooler assemblies and cores covered by this specification are intended for use in military tactical and transport vehicles to cool the engine and transmission oil. Due to the extreme conditions under which military tactical and transport vehicles are required to perform, the cooler assemblies and/or cores covered by this specification are military unique items. Commercial items operating under similar conditions would experience catastrophic failure.

6.2 Acquisition requirements. Acquisition documents should specify the following:

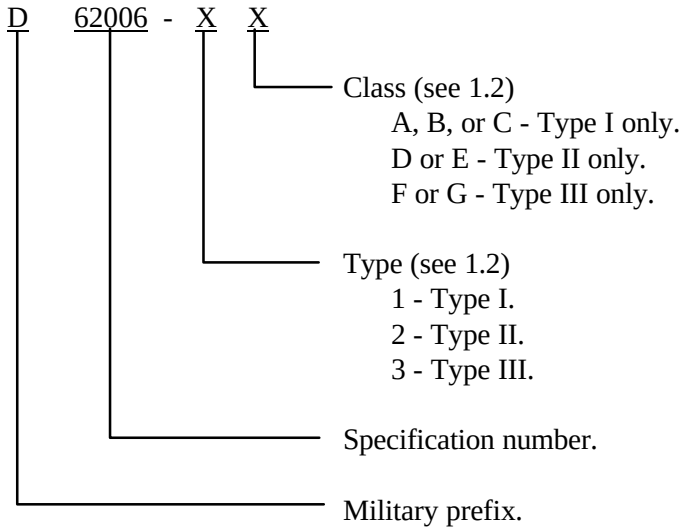
a. Title, number, and date of this specification.
b. Type, PIN, and quantity of coolers required (see 1.2 and 6.5).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. If first article testing is required (see 3.1 and 4.1.1).
e. If identification marking is other than as specified (see 3.5.3).
f. Specification of conformance testing (see 4.1.2).
g. Specification of sampling plan (see 4.1.3).
h. If test conditions are other than as specified (see 4.3.2).
i. Packaging requirements (see 5.1).

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6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table III and their designated frequency based on a risk assessment for the procurement.

6.5 Part or identifying number (PIN). The PINs to be used for coolers acquired to this specification are created as follows:



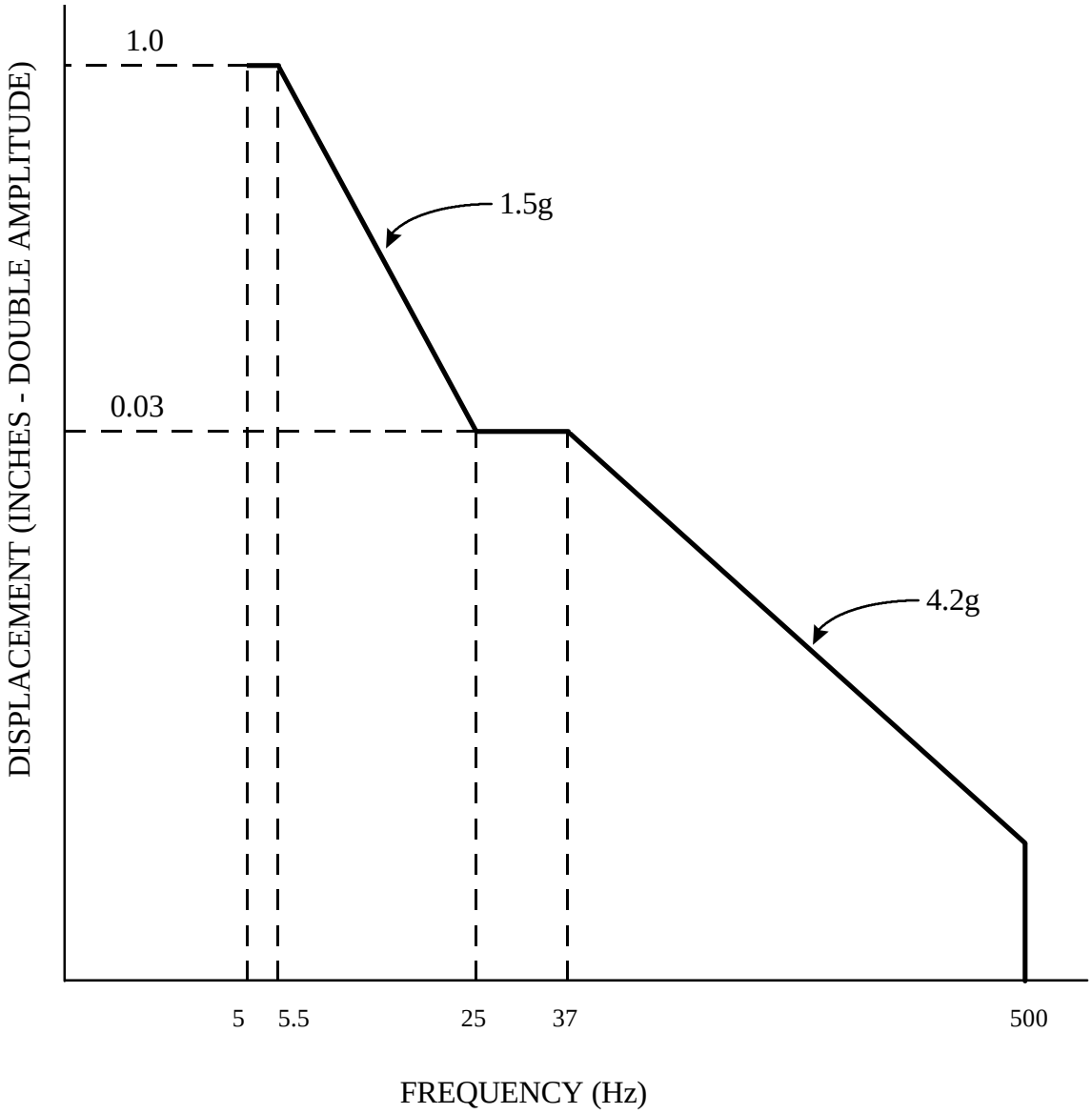
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6.6 Subject term (key word) listing.

Core
Heat rejection
Hydraulic pressure
Hydrostatic pressure
Pneumatic pressure

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensive nature of the changes.

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NOTES:
1. Acceleration levels; ± gravitational units (g) (peak).
2. The curve shall be extended to 2 Hz when test item resonances below 5 Hz are expected.

FIGURE 1. Vibration test curve.

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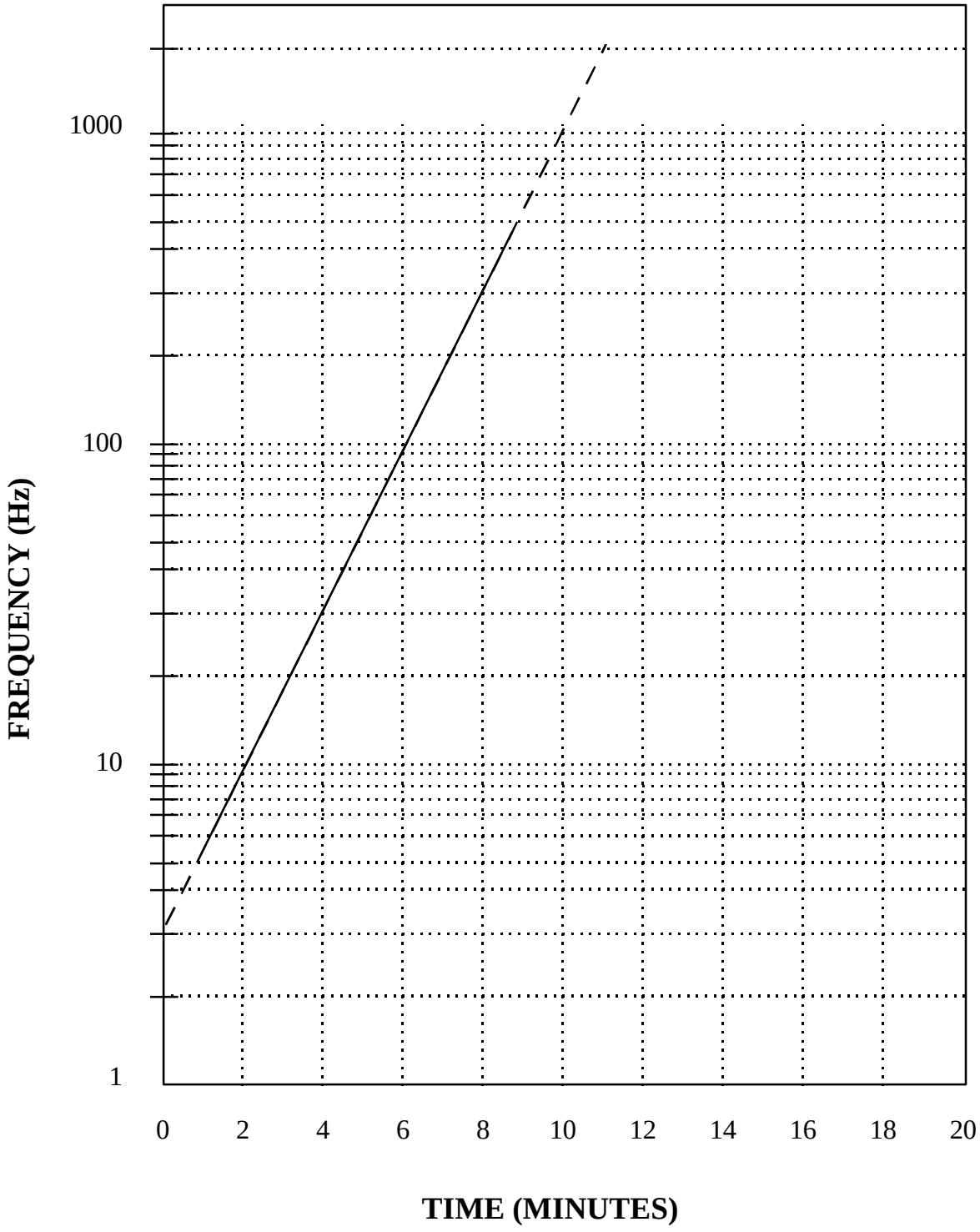


FIGURE 2. Logarithmic sweep.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-62006C(AT)	**2. DOCUMENT DATE *(YYMMDD)*** 1 July 1998

3. DOCUMENT TITLE

Cooler, Oil: Transmission and Engine.

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** **(810) 574-8745** **786-8745**
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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